

Santhosh Naik Dungavath

santhoshn642003@gmail.com — [LinkedIn](#) — [GitHub](#)

Education

New Jersey Institute of Technology — M.S. Artificial Intelligence *GPA: 3.58 / 4.0* *May 2027*

Indian Institute of Information Technology, Dharwad — B.Tech, Computer Science & Engineering *May 2025*

Experience

Research Intern, Dept. of CSE — Medical Speech Recognition *IIIT Dharwad Jun – Nov 2024*

- Developed self-training and LLaMA-based best-hypothesis selection (top- n) to improve Whisper ASR on medical speech, reducing Word Error Rate by 1–2%; built a specialized OpenFDA dataset for evaluation. (Published, IEEE ICEI 2024.)

Research Intern, Dept. of CSE — Low-Resource Speech Translation *IIIT Dharwad Jun – Nov 2024*

- Co-developed a direct speech-to-speech translation system for the low-resource Lambani language using discrete speech units, alongside a faculty advisor and PhD student.

Subject Matter Expert — Computer Science Engineering *Chegg Inc. (Remote) Apr – Dec 2023*

- Authored high-quality solutions and clear explanations to student queries across CSE topics, providing academic support to enhance the learning experience.

Student Peer Educator, Campus Wellness Center *NJIT Nov 2025 – Present*

- Promote MenB and HPV vaccine awareness among students; collaborate with peer educators to design gamified initiatives that increase engagement and health awareness on campus.

Research Projects

Multimodal TEC Forecasting with a Solar Foundation Model *NJIT*

- Built a multimodal deep learning pipeline predicting global ionospheric TEC during geomagnetic storms as residual/anomaly prediction over a frozen NASA/IBM Surya foundation model; reached validation RMSE 6.35 TECU, beating persistence. Trained on Google Colab (A100/H100) and NJIT Wulver HPC (Slurm).

When Context Hurts: Detecting Psychological Defense Mechanisms in Support Conversations *NJIT — DS680*

- Fine-tuned RoBERTa to 0.2604 macro F1 on PSYDEFCONV (BioNLP 2026 shared task), beating MentalBERT; showed a single supporter turn collapses masking-type detection ($p = 0.031$); ran SHAP and context-window ablations.

EmoMusicTransformer: Emotion-Conditioned MIDI Music Generation *NJIT — DS677*

- Built a GPT-style Transformer decoder conditioned on emotion over the EMOPIA dataset (val perplexity 1.89), with per-layer FiLM conditioning and an inference-time emotion-trajectory mechanism for multi-emotion generation.

Publication

Dungavath Santhosh Naik, Nitya Tiwari, K. S. Nataraj. “Self-Training and Error Correction using Large Language Models for Medical Speech Recognition.” *2024 IEEE Conference on Engineering Informatics (ICEI)*.

Skills

Languages: Python, C++, C, SQL **ML/NLP:** PyTorch, Hugging Face Transformers, TensorFlow, Keras, scikit-learn, NumPy, Pandas

Areas: NLP, Deep Learning, Speech Processing, Multimodal Learning, Foundation Models **Infra:** Slurm/HPC, Docker, AWS

Leadership & Service

- **Secretary & PR Head, Sanskar** (NJIT, 2025–2026): Led communications and event logistics for a 100+ member club (Garba Night, Diwali Mela, campus badminton tournament, Game Night).
- **Department Representative – Data Science, GSA** (NJIT, May 2026–Present): Liaison between Data Science graduate students and the GSA.